Software Requirements Specification

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1. Introduction

FixMyBD is a Citizen Public Issue Reporting & Tracking Platform that allows citizens to report public issues in their local areas and track the progress of their reports. Citizens can submit issue details, category, location and photo evidence through the platform.

The system allows responsible authorities to review reports, provide responses, and update issue progress. It also allows citizens to follow these updates and provide feedback. The main goal of FixMyBD is to make public issues more visible, organized and trackable.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly describe the requirements of the FixMyBD system.

This document explains what the system will do, who will use it, and how the main parts of the system are expected to work. It provides a common understanding of the project requirements for the development and testing of the system.

The SRS will also help the project team stay within the defined project scope and ensure that the planned requirements are properly considered during development.

1.2 Intended Audience

This SRS document is intended for the people involved in the development, supervision, testing, and use of the FixMyBD system.

The intended audience includes:

- **Project Developers:** To understand the system requirements and develop the planned features.
- **Project Supervisor/Teacher:** To review the project requirements, objectives and scope.
- **Testers:** To check whether the system works according to the specified requirements.
- **Citizens:** To understand the main functions available for reporting and tracking public issues.
- **Authorities/Departments:** To understand how they can review reports, respond to citizens and update issue progress.
- **System Administrator:** To understand the requirements for managing users, authorities, categories, locations, and reports.

1.3 Intended Use

FixMyBD is intended to be used as a centralized platform for reporting and tracking public issues.

Citizens will use the system to:

- Register and log in.
- Report public issues.
- Select an issue category and location.
- Add a description and photo evidence.
- Track the status and progress of their reports.
- View authority responses and updates.
- Support existing issues.
- Provide comments and feedback.

Authorities/Departments will use the system to:

- View submitted reports.
- Review and assess reported issues.
- Respond to citizens.
- Update the status of reports.
- Add progress information.
- Upload supporting evidence.
- Respond to citizen feedback.

System Administrators will use the system to:

- Manage users and authorities.
- Manage issue categories and locations.
- Monitor submitted reports.
- Moderate inappropriate reports.
- View system analytics.
- Manage system notifications.

1.4 Product Scope

FixMyBD will be developed as a web-based platform for reporting, tracking and managing public issues.

The system will provide the following major functions:

Citizen Features

- Registration and Login
- Public Issue Reporting
- Issue Category Selection
- Location Selection
- Photo Upload
- Report Tracking
- Progress Timeline
- Authority Updates
- Issue Support/Upvote
- Comments
- Citizen Feedback
- Notifications

The location structure will follow:

Division → District → Upazila → Union → Ward

Authority Features

- View submitted reports
- Review and assess reports
- Respond to citizens
- Update report status
- Add progress information
- Upload supporting evidence
- Respond to citizen feedback

The report status will generally follow:

Submitted → Acknowledged → Under Assessment → Action Planned → In Progress → Completed → Citizen Feedback

The system will also support **On Hold, Rejected and Reopened** statuses when required.

Admin Features

- User Management
- Authority Management
- Category Management
- Location Management
- Report Monitoring
- Content Moderation
- Dashboard and Basic Analytics
- System Notifications

FixMyBD is an **academic prototype**. The project will focus on reporting, communication, tracking and feedback. It will not directly integrate with government systems or perform the actual public service work. The actual resolution of an issue will remain the responsibility of the relevant authority.

1.5 Risk Definitions

The following risks may affect the FixMyBD system:

Risk	Definition
False or Inappropriate Reports	Citizens may submit incorrect, misleading, or inappropriate reports, which can affect the quality of system data.
Incorrect Location Information	A citizen may provide an incorrect location, making it difficult for the responsible authority to identify the actual issue.
Data Privacy	Personal information and uploaded evidence need to be handled carefully to protect citizen privacy.
Data Security	Unauthorized access to system information may affect the security and reliability of stored data.

Risk	Definition
Delayed Authority Response	Authorities may take time to review or respond to a report, which can delay progress updates for citizens.
System Availability	Server or network problems may temporarily prevent citizens or authorities from using the system.
System Performance	A large number of users and reports may affect system performance as the system grows.
Project Time Limitation	Since FixMyBD is an academic project with a limited development period, some advanced features may not be included in the initial version.

2. Overall Description

The following sections provide an overall description of the FixMyBD system by identifying its major user groups, their needs, the environment in which the system will operate, and the factors that may affect its development and operation. The User Classes and Characteristics section describes the different types of users, including citizens, authorities and system administrators, along with their responsibilities and expected level of interaction with the system. The User Needs section explains the primary requirements and expectations of each user group. The Operating Environment describes the hardware, software, technologies, and external services required for the system to function. The Constraints section identifies the technical, project, and operational limitations that may influence the development of FixMyBD. Finally, the Assumptions and Dependencies section describes the conditions assumed during development and the external technologies or services on which the system depends. Together, these sections establish the overall context and operational characteristics of the FixMyBD platform.

2.1 User Classes and Characteristics

The FixMyBD system has three main user classes: **Citizens, Authorities/Departments and System Administrators**.

- **Citizens:** Citizens are the primary users of FixMyBD. They can register and log in to the system, report public issues, select the appropriate category and location, upload photos as evidence, and track the progress of their submitted reports. They can also view authority responses, support existing issues, comment on reports and provide feedback regarding the resolution of an issue. Citizens are expected to have basic knowledge of using web applications.
- **Authorities/Departments:** Authorities are responsible for reviewing and managing public issue reports submitted by citizens. They can view submitted reports, assess or

verify reported information, respond to citizens, update report statuses, add progress information and upload supporting evidence. Authorities require basic computer and web application knowledge and access to the reports relevant to their department or area.

- **System Administrators:** Administrators have the highest level of system access. They manage users, authorities, issue categories and location information. They can monitor reports, moderate inappropriate content, manage system notifications and access dashboards and analytics. Administrators are expected to have sufficient technical and administrative knowledge to operate and maintain the platform.

2.2 User Needs

The FixMyBD system is designed to address the needs of its different user groups.

Citizen Needs

- An easy and structured method for reporting public issues.
- The ability to provide issue descriptions, categories, locations and photographic evidence.
- The ability to track the current status and progress of submitted reports.
- Visibility into authority responses and updates.
- The ability to support and comment on existing issues.
- A mechanism for providing feedback about issue resolution.
- Notifications about important changes to submitted reports.

Authority Needs

- A centralized dashboard for viewing submitted public issue reports.
- Access to relevant issue details, locations, categories and evidence.
- The ability to review and assess reported issues.
- The ability to communicate responses to citizens.
- The ability to update issue status throughout the resolution process.
- The ability to add progress information and supporting evidence.
- Better visibility of pending and completed issues.

Administrator Needs

- User and authority management.

- Management of issue categories and location hierarchy.
- Monitoring and moderation of submitted reports.
- Management of inappropriate or invalid content.
- Access to basic system analytics and statistics.
- Ability to maintain the overall organization and reliability of the platform.

2.3 Operating Environment

FixMyBD is proposed as a **web-based application** that can be accessed through modern web browsers on computers and other supported devices.

The proposed operating environment consists of:

- **Frontend:** HTML, CSS and JavaScript.
- **Backend:** Node.js.
- **Database:** MongoDB.
- **Map/Location Service:** OpenStreetMap.
- **Development Environment:** Visual Studio Code.
- **Version Control:** Git and GitHub.
- **Client Environment:** A modern web browser with an active internet connection.
- **Server Environment:** A server capable of running the Node.js backend and connecting to the MongoDB database.

The system should support location-based issue reporting using the defined hierarchy:

Division → District → Upazila → Union → Ward.

2.4 Constraints

The development and operation of FixMyBD are subject to the following constraints:

- The project will be developed using the proposed technology stack of **HTML, CSS, JavaScript, Node.js and MongoDB**.
- The project is intended to be completed within the available **course/project development period**.
- The initial version is an **MVP/prototype** rather than a fully integrated government service.

- FixMyBD will not directly control or perform actual government/public-service operations. Authorities are responsible for taking real-world action on reported issues.
- The system depends on the availability of an internet connection for web-based access and communication with external location services.
- The system's location functionality depends on the availability and capabilities of **OpenStreetMap**.
- The accuracy of issue information depends partly on the information and evidence submitted by citizens.
- Authority response times cannot be guaranteed by the system because actual responses depend on the responsible authorities.
- The initial system may provide basic analytics rather than advanced government-level data analysis.
- Storage, server resources, development time and project scope may limit the number of features implemented in the initial release.

2.5. Assumptions

- Users will have access to a device capable of running a modern web browser.
- Citizens will provide reasonably accurate information when submitting reports.
- Citizens will select appropriate categories and locations for their reported issues.
- Citizens may provide photographs or other supporting evidence when available.
- Authorities using the platform will have appropriate access to the reports assigned to them.
- Authorities will update report statuses and provide progress information when appropriate.
- Administrators will maintain valid user, authority, category, and location information.
- Users will have basic knowledge of web applications.
- The system will be used primarily for reporting and tracking public issues within the defined project scope.

3. Requirements

3.1 Functional Requirements

FR-01: User Registration

User Story:

As a citizen, I want to create an account so that I can report and track public issues.

Confirmation:

The system shall allow a citizen to register using the required personal information and create an account successfully.

FR-02: User Login

User Story:

As a registered user, I want to log in to the system so that I can access my account and available features.

Confirmation:

The system shall authenticate valid login credentials and provide access according to the user's role.

FR-03: Report Public Issue

User Story:

As a citizen, I want to report a public issue so that the responsible authority can review it.

Confirmation:

The system shall allow citizens to submit an issue with a title, description, category, location and supporting photo/evidence.

FR-04: Select Issue Location

User Story:

As a citizen, I want to specify the location of an issue so that the authority can identify where the problem exists.

Confirmation:

The system shall allow the citizen to select a location using the hierarchy Division → District → Upazila → Union → Ward.

FR-05: View Submitted Reports

User Story:

As a citizen, I want to view my submitted reports so that I can monitor the issues I have reported.

Confirmation:

The system shall display the citizen's submitted reports with relevant report information and current status.

FR-06: Track Report Status

User Story:

As a citizen, I want to track the status of my report so that I know how the authority is handling the issue.

Confirmation:

The system shall display the current status and progress timeline of a submitted report.

FR-07: Authority Review

User Story:

As an authority user, I want to review submitted reports so that I can assess the reported public issues.

Confirmation:

The system shall allow authorized authority users to view and assess submitted reports.

FR-08: Authority Response

User Story:

As an authority user, I want to respond to a citizen's report so that the citizen can know what action is being taken.

Confirmation:

The system shall allow authority users to provide responses and progress information for reports.

FR-09: Update Report Status

User Story:

As an authority user, I want to update the status of a report so that citizens can track its progress.

Confirmation:

The system shall allow authorized authority users to update a report through the defined statuses:

Submitted → Acknowledged → Under Assessment → Action Planned → In Progress → Completed

The system shall also support On Hold, Rejected and Reopened where applicable.

FR-10: Citizen Feedback

User Story:

As a citizen, I want to provide feedback on the authority's response so that I can indicate whether the issue has been properly addressed.

Confirmation:

The system shall allow citizens to provide feedback after receiving an authority update.

FR-11: Comment on Issue

User Story:

As a citizen, I want to comment on an issue so that I can provide additional information or observations.

Confirmation:

The system shall allow authorized users to submit comments on eligible reports.

FR-12: Notification

User Story:

As a citizen, I want to receive notifications about my report so that I know when important changes occur.

Confirmation:

The system shall notify users when a report is submitted, its status changes, an authority responds, or feedback is requested.

FR-13: Administrator User Management

User Story:

As an administrator, I want to manage users so that I can maintain proper access to the system.

Confirmation:

The system shall allow authorized administrators to view, manage, and control user accounts.

FR-14: Authority Management

User Story:

As an administrator, I want to manage authorities and departments so that reports can be handled by the appropriate authority.

Confirmation:

The system shall allow administrators to add, update, and manage authority information.

FR-15: Report Monitoring

User Story:

As an administrator, I want to monitor submitted reports so that I can oversee the overall system activity.

Confirmation:

The system shall allow administrators to view and monitor reports and their current statuses.

FR-16: Content Moderation

User Story:

As an administrator, I want to moderate inappropriate reports and content so that the platform remains useful and appropriate.

Confirmation:

The system shall allow authorized administrators to review and take appropriate action on inappropriate or invalid content.

FR-17: Search and Filtering

User Story:

As a user, I want to search and filter reports so that I can find relevant public issues easily.

Confirmation:

The system shall allow reports to be filtered based on available information such as category, location, and status.

FR-18: Dashboard and Analytics

User Story:

As an administrator, I want to view issue statistics so that I can understand reported problems by area and category.

Confirmation:

The system shall provide dashboard information and basic statistics based on available report data.

3.2 Non-Functional Requirements

NFR-01: Performance

- The system should respond to normal user requests within an acceptable time.
- The report submission process should complete without unnecessary delay.
- Dashboard and report information should load efficiently.

NFR-02: Security

- The system shall authenticate users before allowing access to protected features.
- The system shall enforce role-based access for Citizens, Authorities, and Administrators.
- User passwords shall be stored securely.
- Unauthorized users shall not be allowed to access restricted information or functions.

NFR-03: Quality

- The system should provide a simple and understandable interface.
- Citizens should be able to submit an issue without requiring advanced technical knowledge.
- Forms should provide clear instructions and validation messages.
- Report status and authority updates should be presented clearly.

NFR-04: Availability

- The system should be available to users during normal operating periods.
- The system should recover properly after an unexpected system failure.

NFR-05: Maintainability

- The system should be developed using a modular structure.
- The source code should be organized so that future modifications can be made easily.
- Git and GitHub should be used for version control.

NFR-06: Scalability

- The system should be designed to accommodate an increasing number of users and reports.
- New categories, locations, and authorities should be addable without major changes to the system.

NFR-07: Data Integrity

- The system shall validate required information before accepting a report.
- Each submitted report shall have a unique identifier.
- Report status and progress information shall be stored consistently.

NFR-08: Safety

- The system should prevent invalid or inappropriate user input from negatively affecting normal system operation.
- Uploaded evidence should be validated before being accepted by the system.